सत्यं विश्वमानम्

WEST BENGAL STATE UNIVERSITY

B.Sc. Major 2nd Semester Examination, 2025

CMSDSC202T-COMPUTER SCIENCE (MAJOR)

Time Allotted: 2 Hours

Full Marks: 50

The figures in the margin indicate full marks.
Candidates should answer in their own words and adhere to the word limit as practicable.
All symbols are of usual significance.

GROUP-A

1. Answer any *five* questions from the following: $2\times5=10$

(a) Distinguish between linear and non linear data structure.

(b) What is the difference between a stack and a queue?

(c) State two applications of linked lists.

(d) What is the worst-case time complexity of quicksort and why?

(e) Differentiate between binary tree and binary search tree.

(f) What is the purpose of a hash function in hashing?

(g) Why is a circular queue better than a linear queue?

(h) How does AVL tree differ from a Binary Search Tree?

GROUP-B

Answer any *five* questions from the following $8\times5=40$

2. (a) Draw the internal memory representation of the binary tree of **Fig 1** using (i) sequential and (ii) linked representation. 3+3+2

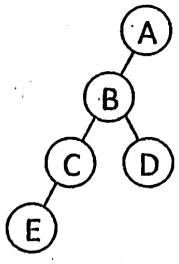

Fig 1

(b) Suppose a binary tree T is traversed in post-order and in-order respectively and the sequences of the nodes are given by:

Post-order:	H	D	I	E	B	J	F	K	L	G	C	A
In-order:	H	D	B	I	E	A	F	J	C	K	G	L

Construct the tree showing the intermediate steps.

(c) What is the number of all possible distinct binary trees of 4 nodes?

3. (a) How will you represent the polynomial $4x^3 - 10x^2 + 3$ using linked list? 2+4+2
   (b) Write an algorithm to reverse the direction of links of a given singly linked list.
   (c) Write an algorithm to delete an element from the beginning of a doubly linked list.

4. (a) Write a C++ program to implement Insertion sort. 4+4
   (b) Calculate the number of comparisons and the number of swapping required for the following sequence to sort in ascending order using the above algorithm:

   80, 15, 37, 92, 12, 56, 44, 63

5. (a) Write an algorithm to find inorder predecessor of an element in a Binary Search Tree. 4+4
   (b) Find the address of a [5] [6] using (i) row major (ii) Column major representation where base address of the array is 1000 and word size = 2. Consider array size as 10×15.

6. Write an algorithm to convert any infix expression to prefix expressions. Using this algorithm convert the following infix expression to prefix expression. (Showing the content of stack in every step). 5+3

$$(a + b \wedge c \wedge d) * (e + f / d)$$

7. (a) Given a hash table of size 10 and keys: 23, 44, 39, 55, 7, 79. Insert using modulo 10 and linear probing. Show the final table. 3+3+2
   (b) Compare the three collision resolution techniques: chaining, linear probing, and quadratic probing with examples.
   (c) What is the load factor in a hash table? Why is it important?

8. (a) Prove that $n_0 = n_2 + 1$, where n_0 is the number of leaf vertices and n_2 is the number of nodes of degree 2 of a non-empty binary tree. 4+4
   (b) What is priority queue? How priority queue can be implemented?

9. Write short notes on: (any *two*) 4+4
   (a) Sparse matrix
   (b) Abstract Data Type
   (c) AVL Tree.

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